Foundational Concepts of Analytics

1) Business Analytics

It is a term that is used to solve business problems and plan for the future

E.g We can look at staffing or store stock and determine what can be done to improve sales and how many employee we can employee.

2) Data Analytics

This is a technical way of cleaning and processing data, in order to give business clear insights of the running of the business.

E.g Taking a messy spreadsheet of 8000 coffee sales and organizing it so it can intrepretted or understood by the business owners.

3) Business Intelligence BI

Tools and systems that turn data into easy to read reports and dashboard. Mainly focusing on the current state of the business.

E.g A live app dash that shows live graph of how many lattes are sold today compared to yesterday.

4) Data Driven Decision Making DDDM

A management approach where decisions are validated with rather than intution or observation.

E.g Instead of thinking people like blueberry muffin you check the data see they sell out by 10:00am, and decide to bake twice as many.

5) Big data

Too large or complex for traditional data processing software, which require special tools to process.

E.g Netflix processing billions of play and pause events every day from millions of users simultaneously.

6) The four stage of analysis

- Descriptive - What happened
- Diagnostic - Why did it happen
- Predictive - What will happen
- Prescriptive - What should we do

6) Descriptive Analytics

Look at past to summarize what has already occured.

E.g A report showing that sales went up by 20% last month.

7) Diagnostic Analytics

Digging deeper into data to find the root cause of a trend.

E.g You realize sales went up because there was local festival held outside your shop that week.

8) Predictive Analytics

Using historical patterns to forecast future outcomes.

E.g Based on season you predict that you will sell 60% more hot chocolate once temperature drops 10°C.

9) Prescriptive Analystics

It suggests specific action to reach a goal.

E.g To maximize profit this weekend, schedule two extra baristas and run a 10% discount on iced coffee between 2pm and 4pm.

10) Data Literacy
The ability to read and understand data, analyze and argue with data.
E.g When a supplier tells you a new bean is performing great, you known how to ask for the specific sales growth percentages and compare them to your current bean before buying.

Data, Measurement and Reporting

11) Structured Data:
This is organised data
E.g Excel table sheets.

12) Unstructured data
The data is not organised, without format.
E.g Music ~~video~~ video, phone calls

13) Datawarehouse:
Central data store
E.g Store collect data from different sources and it's stored in EDW

14) Database
Is a place where store data
E.g It is a way of storing data in tables like excel sheet that linked to each other using keys.

15) Data Mining
A process of extracting data.
E.g Using software to discover that people who buy blueberry muffins almost always also buy a dark roast coffee, even if they don't buy at the same transaction.

16) Data Visualization

Being able to convert tables into picture, so the human brain can understand them instantly.

E.g Showing the daily sales such as a line, bar, pie graph.

17) Dashboard

A screen display with most important data visualization in real time.

E.g The tablet on your counter showing today's total sales, current stock levels and the number of customers currently in the shop.

18) Metric

Any number that tracks a specific business activity.

E.g Number of cups sold or Total cost of milk per week.

19) Key Performance Indicator KPI

A form that tells the growth of a business; succeeding or failing.

E.g While number of cups sold is a metric, your KPI might be profit Margin percentage, because that is the number that dictate if you can afford to pay rent.

20) Benchmark

It is a standard number used to compare your performance against.

E.g If your shop has a 10% margin profit, you look up the industry average for coffee shops. If the industry average is 15%, you know you have room to improve.

21) Segmentation

Dividing your total data into smaller, meaningful groups.

E.g Splitting your customers list into daily commuters, who buy coffee at 7am and Weekend tourist who buy at 11am.

Customer and Sales KPIs

22) Conversion Rate

The percentage of people who do what you want them to do.

E.g. If 100 people visit your website and 5 of them buy a product your conversion rate is 5%.

23) Customers Acquistion Cost

The total money you spend on marketing and sales to get one new customer.

E.g. You spend R1000 on social media ads in a month and gain 20 new customers. Your CAC is R50 per person.

24) Customer Lifetime Value (CLV or LTV)

The total profit you expect to earn from a single customer over the entire time they remain with your business

E.g. A customer visits your store once a week and spends R100 profit for 2 years. Their LTV is roughly R10,400.

25) Churn Rate

The percentage of customers who stop doing business with you over a specific time.

E.g. If you start the month with 100 subscribers and 5 cancel your churn rate is 5%.

26) Customer Retention Rate

The percentage of customers who stay with you. This is simply the inverse of your churn rate.

E.g. If your churn rate is 5%, your retention rate is 95%

27) Net promoter Score (NPS)

A loyal metric based on one simple question. On a scale of 0 to 10 how likely are you to recommend us to a fried.

E.g People who answer 9-10 are Promoter' while those who answer 0-6 are ~~deat~~ detractors. A high NPS suggests your custom are doing your marketing for you.

28) Average Order Value (AOV)

The average dollar amount spent each time a customer complete an order.

E.g You have two sales: one for $100 and one for $300. Your AOV is $200

29) Sales Funnel

The visual map of a customer's journey from being a stranger (Awareness) to becoming a loyal fan (Purchase/Advocacy)

E.g Top: Wide: A person sees you ad (awareness)
Middle: They visit your site and read about your product (interest)
Bottom: They add the item to their cart and pay (Purchase)

30) Marketing Campaign

A set of coordinated activities designed to achieve a specific goal over a set period.

E.g A spring sale campaign running for two weeks across email and facebook.

31) Incremental Revenue

The extra money you make during the campaign, beyond what you would have made anyway.

E.g You normally sell 100 units a week. During your campaign you sell 150. That 50 unit increase is your incremental revenue.

32) Post-campaign analysis

You look at the data to see what worked and what didn't.

E.g Realizing your instagram ads worked great, but email blasts didn't lead to many clicks.

33) Return of Investment ROI

The ultimate measure of profitability.

E.g You spent R1000 on ads and made R3000 in profit. Your ROI is 200%.

34) Return on Ad Spend (ROAS)

Similar to ROI, but it measures gross revenue against the cost of the ads specifically.

E.g You spend R1000 on ads and generate R5000 in total sales. Your ROAS is 5:1.

35) Click through Rate CTR

The percentage of people who see your ad/link and actually click on it.

E.g 1000 people see your ad, and 20 click it. Your CTR is 2%.

36) Bounce Rate

The percentage of visitors who land on your page and leave immediately without clicking anything else

E.g Someone clicks your ad, sees the landing page and close the tab within 3 seconds. That's a bounce

37) AB Testing.

Running two versions of something to see which performs better

E.g Showing half your visitors a "Buy Now" button in blue and the other half in red to see which color get more clicks.

38) Lead generation

The process of attracting people to your business and getting their contact info (name, email) so you can sell to them later.

E.g Offering a free "How to guide" in exchange for a customer's email address.

39) Marketing Attribution

The logic used to figure out which touchpoint deserves credit for a sale.

E.g A customer saw you TikTok then searched for you on google then clicked an email link to buy. Attribution decides if TikTok google or email gets the "win".